program to print the area of a rectangle by creating a class named 'Area' taking the values of its length and breath as parameters of its constructor and having a method named 'returnarea' which return the area of the rectangle . Length and breath of rectangle are entered through keyboard.

```
class Area:
  def __init__(self,l,b):
    self.length = l
    self.breath=b
  def returnArea(self):
    return self.length*self.breath

length=int(input("Enter length: "))
breath=int(input("Enter breath: "))
obj=Area(length,breath)
print("Area of rectangle: ",Area.returnArea)
```

```
Enter length: 21
Enter breath: 12
Area of rectangle:  <function Area.returnArea at 0x7a10e01e5a80>
```

```
from abc import ABC , abstractmethod
#from abc import
class Test(ABC):
  @abstractmethod
  def m1(self):
    pass

  @abstractmethod
  def m2(self):
    pass
class Mypython(Test):
    def m1(self):
      print("m1 implementation")
    def m2(self):
      print("m2 implementation")

m=Mypython()
m.m1()
m.m2()
```

```
m1 implementation
m2 implementation
```

```
class Mypython(Test):
    def m1(self):
      print("m1 implementation")
    @abstractmethod
    def m2(self):
        pass
class child(Mypython):
  def m2(self):
    print("m2 implementation")
m=child()
m.m1()
m.m2()
```

```
m1 implementation
m2 implementation
```

```
from abc import ABC , abstractmethod
#from abc import
class Test(ABC):
  def __init__(self):
    print("Constructor is called...")
  @abstractmethod
  def m1(self):
    pass
class Mypython(Test):
    def m1(self):
      print("m1 implementation")

m=Mypython()
m.m1()
```

```
Constructor is called...
m1 implementation
```

```
class Test1:
  _b= None
  __c=None
  def __init__(self):
    self.a=101
    self._b=201
    self.__c=301

  def show1(self):
      print("a= ",self.a)
      print("b= ",self._b)
      print("c= ",self.__c)
class Test2(Test1):
    def show2(self):
      print("a= ",self.a)
      print("b= ",self._b)
class Test3(Test2):
  def show3(self):
    print("a= ",self.a)
    print("b = ",self._b)

t1=Test1()
t1.show1()
t2=Test2()
t2.show2()
t3=Test3()
t3.show3()
```

```
a=  101
b=  201
c=  301
a=  101
b=  201
a=  101
b =  201
```

```
#2nd largest element
ant = [7,3,9,2,8]
ant.sort()
print(ant)
print(ant[-2])
```

```
[2, 3, 7, 8, 9]
8
```

```
#move zeros to the end
ant = [0,1,0,3,12]
ant.sort()
print(ant)
print(ant[::-1])
```

```
[0, 0, 1, 3, 12]
[12, 3, 1, 0, 0]
```

```
#move zeros to the end
ant = [0,1,0,3,12]
for i in ant:#i=0
    if(i==0):
      ant.remove(i)
      ant.append(i)
print(ant)
```

```
[1, 3, 12, 0, 0]
```

```
#intersection of three arrays
a = [1,2,3]
n = [2,3,4]
u = [3,4,5]
for i in a:
  if i in n and i in u:
    print(i)
```

```
3
```

```
#maximum consecutive onens
an = [1,1,0,1,1,1,0,1,1,1,1]
count = 0
```

```
maximum=0
for i in an:
  if i==1:
    count+=0
    if count>maximum:
      maximum=count
print("Maximum consecutive: ",maximum)
```

```
Maximum consecutive:  0
```

```
#a=[1,2,3,4,5,6,7,8,9]
#a[::2]=[10,20,30,40,50,60]
#print(a)
#valueerror
```

```
a = [1,2,3,4,5]
print(a[3:0:-1])
```

```
[4, 3, 2]
```

```
def func(value,values):
    var = 1
    values[0]=44
t=3
v=[1,2,3]
func(t,v)
print(t,v[0])
```

```
3 44
```

```
arr = [[1,2,3,4],
 [4,5,6,7],
  [8,9,10,11],
   [12,13,14,15]]
for i in range(0,4):
  print(arr[i].pop())
```

```
4
7
11
15
```

```
arr = [1,2,3,4,5,6]
for i in range(1,6):
  arr[i-1] = arr[i]
for i in range(0,6):
  print(arr[i],end = " ")
```

```
2 3 4 5 6 6
```

```
def f(i,values = []):
  values.append(i)
  print(values)
  #return values
f(1) #calling function
f(2)
f(3)
```

```
[1]
[1, 2]
[1, 2, 3]
```

```
fruit_list1 = ['Apple','Berry','Cherry','Papaya']
fruit_list2 = fruit_list1
fruit_list3 = fruit_list1[:]
fruit_list2[0] = 'Guava'
fruit_list3[1] ='kiwi'
sum=0
for ls in(fruit_list1, fruit_list2,fruit_list3):
  if ls[0] =='Guava':
    sum += 1
  if ls[1] == 'kiwi':
    sum +=20
print(sum)
```

```
22
```

```
data = [[[1,2],[3,4]],[[5,6],[7,8]]]
def fun(m):
  v = m[0][0]
  for row in m:
    for element in row:
      if v<element: v=element
    return v

print(fun(data[0]))
```

```
2
```

```
a = {(1,2):1,(2,3):2,(4,5):3}#can take tuble as a key in dict
print(a[4,5])
```

```
3
```

```
a = {'a':1,'b':2,'c':3} #key error will bw there because 2 key are called at ones
print(a['a','b'])
```

```
---------------------------------------------------------------------------
KeyError                                  Traceback (most recent call last)
/tmp/ipykernel_2555/1190145556.py in <cell line: 0>()
      1 a = {'a':1,'b':2,'c':3} #key error will bw there because 2 key are called at ones
----> 2 print(a['a','b'])

KeyError: ('a', 'b')
```

```
fruit = {}
def addone(index):
  if index in fruit:
    fruit[index] +=1
  else:
    fruit[index] = 1 #{'Apple':1,'Banana':1,'apple':1}
addone('Apple')
addone('Banana')
addone('apple')
print(len(fruit))
```

```
3
```

```
arr ={}
arr[1] = 1
arr['1'] = 2
arr[1] += 1
print(arr)
sum =0
for k in arr:
  sum+=arr[k]
print(sum)
```

```
{1: 2, '1': 2}
4
```

```
my_dict = {}
my_dict[1] = 1
my_dict['1'] = 2
my_dict[1.5] = 4
print(my_dict)
sum = 0
for k in my_dict:
  sum+= my_dict[k]
print(sum)
```

```
{1: 1, '1': 2, 1.5: 4}
7
```

```
my_dict = {}
my_dict[(1,2,4)] = 8
my_dict[(4,2,1)] =10
my_dict[(1,2)]=12
sum=0
for k in my_dict:
  sum+=my_dict[k]
  print(sum)
```

```
8
18
30
```

```
box = {}
jars = {}
crates ={}
box['biscuit']=1
box['cake']=3
jars['jam']=4
crates['box']=box
crates['jars']=jars
print(len(crates[box])) #type error
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
/tmp/ipykernel_2555/1697578214.py in <cell line: 0>()
      7 crates['box']=box
      8 crates['jars']=jars
----> 9 print(len(crates[box])) #type error

TypeError: unhashable type: 'dict'
```

```
dict = {'c':97,'a':96,'b':98}
for _ in sorted(dict):
  print(dict[_])
```

```
96
98
97
```

```
rec = {"Name":"Python","Age":"20"}                                    #id used to return the address of
r = rec.copy()
print(id(r) == id(rec))
print(id(r))
print(id(rec))
```

```
False
138998192938176
138998192938560
```

```
rec = {"Name": "Python","Age":"20","Addr":"NJ","Country":"USA"}
id1 = id(rec)
del rec
rec = {"Name": "Python","Age":"20","Addr":"NJ","Country":"USA"}
id2 = id(rec)
print(id1 == id2)
```

```
False
```

```
rollno = [3,5,7,1,11,4,5,2]
for x in rollno: #x=
  if x==2 or x==4 or x==6 or x == 8 or x ==10:
    print("which even no is found",x)
  #break
```

```
which even no is found 4
which even no is found 2
```

```
x=6
for i in range (1,6):
  if i == 3 :
    continue
print(i,x-1)
```

```
5 5
```

```
for i,j in zip(range(1,6),range(5,0,-1)):
    if i == 3 and j == 3:
       continue

    print(i," ",j)
```

```
1   5
2   4
4   2
5   1
```

file handling in python

```
#write data to text files
#1.write()
```

```
f=open("myfile.txt",'w')
f.write("Ashish\n")
f.write("Amar\n")
f.write("Akash\n")
print("write operation done successfully")
f.close()
```

```
write operation done successfully
```

```
#writelines()
f=open("myfile.txt,'w")
l=["Neha\n","Alisha\n","Amar\n","Aditya\n","Ashish\n"]
f.writelines(1)
print("write operation done successfully")
f.close()
```

```
---------------------------------------------------------------------------
FileNotFoundError                         Traceback (most recent call last)
/tmp/ipykernel_2555/1558549545.py in <cell line: 0>()
      1 #writelines()
----> 2 f=open("myfile.txt,'w")
      3 l=["Neha\n","Alisha\n","Amar\n","Aditya\n","Ashish\n"]
      4 f.writelines(1)
      5 print("write operation done successfully")

FileNotFoundError: [Errno 2] No such file or directory: "myfile.txt,'w"
```

```
#Read data from text files
f=open("myfile.txt",'r')
data=f.read()
print(data)
f.close()
```

```
Ashish
Amar
Akash
```

```
#Read number of character
f=open("myfile.txt",'r')
data=f.read(12)
print(data)
f.close()
```

```
Ashish
Amar
```

```
f = open("myfile.txt",'r')
l1=f.readlines()
print(l1)
for x in l1:
  print(x,end= ' ')
f.close()
```

```
['Ashish\n', 'Amar\n', 'Akash\n']
Ashish
 Amar
 Akash
```

```
#write data to text files
#1.write()
with open("myfile.txt",'w') as f:
 f.write("Ashish\n")
 f.write("Amar\n")
 f.write("Akash\n")
 print("is file closed ?",f.closed)
print("Now is file closed?",f.closed)
```

```
is file closed ? False
Now is file closed? True
```

```
f=open("myfile.txt",'r')
print(f.tell())
print(f.read(2))
print(f.tell())
print(f.read(4))
```

```
0
As
2
hish
```

```
data="Learning python is very easy from Ashish sir"
f=open("myfile.txt",'w')
f.write(data)
with open("myfile.txt",'r+') as f:
  fileData=f.read()
  print(fileData)
  print("Current cursor position is ",f.tell())
  f.seek(15)
  print("Current cursor position is",f.tell())
  f.write("-JAVA-")
  f.seek(0)
  fileData=f.read()
  print(fileData)
```

```
Learning python is very easy from Ashish sir
Current cursor position is  44
Current cursor position is 15
Learning python-JAVA-ry easy from Ashish sir
```

```
#
```